

Data vs Information

1. What is Data?

Definition

Data is a collection of raw, unprocessed facts and figures that may not have meaningful context by themselves.

The word **raw** is the key.

Data can be:

- Numbers
- Text
- Characters
- Images
- Audio
- Video
- Measurements
- Symbols

For example:

75
82
91

These are simply numbers.

If I tell you:

75, 82, 91

you don't necessarily know what they represent.

They could be:

- temperatures
- prices
- student marks

- ages
- product quantities

So they are **data**.

2. Examples of Data

Student data

Name: Ali

Age: 19

Marks: 850

These individual facts are data.

Store data

Rice 250

Milk 180

Eggs 320

Again, these are raw data.

Sensor data

Imagine a temperature sensor producing:

31°C

32°C

33°C

34°C

These measurements are data.

3. What is Information?

Now the important part.

Information is processed, organized, or interpreted data that has meaning and is useful to the user.

In simple words:

Data becomes information when processing gives it meaning.

For example:

Raw data:

75

82

91

Processing:

$(75 + 82 + 91) / 3$

Result:

82.67

If these are a student's marks, we can say:

The student's average mark is 82.67.

Now the data has been processed and interpreted.

That is **information**.

4. The Fundamental Relationship

Remember this:

DATA

↓

PROCESSING

↓

INFORMATION

This is one of the most important models in introductory Computer Science.

5. Data vs Information — Simple Example

Imagine a teacher has these marks:

Ali 75
Ahmed 82
Sara 91

This is **data**.

The computer processes the marks and determines:

Highest mark = 91
Lowest mark = 75
Average = 82.67

These results are **information**.

Why?

Because the raw marks have been processed into something meaningful and useful.

6. Another Example — Shopping

Suppose you have:

Rice = 250
Milk = 180
Eggs = 320
Bread = 150

These are **data**.

The computer calculates:

$250 + 180 + 320 + 150$
 $= 900$

Now:

Total bill = Rs. 900

is information.

So:

Raw prices

↓
Addition
↓
Total bill

7. Data Can Be Different Forms

Data isn't only numbers.

Textual data

Taifoor
Computer Science
Pakistan

Numerical data

18
850
95.5

Image data

A photograph is data.

Audio data

A recorded voice is data.

Video data

A video file is data.

Sensor data

Temperature = 35°C
Pressure = 1013 hPa

All of these can be processed by computers.

8. A Very Important Point

Don't think:

"Data is always meaningless."

That's not exactly correct.

Data can have some meaning to a human, but **information is data that has been processed/organized so that it provides useful meaning in a particular context.**

For example:

95

is data.

But:

Ahmed scored 95% in Mathematics.

provides much more useful context.

9. Data vs Information

Data	Information
Raw facts	Processed/organized data
Usually lacks context	Has meaningful context
Input to processing	Result of processing
May not be directly useful	Usually useful for understanding/decision-making
Can be numbers, text, images, etc.	Gives meaning to processed data

Remember:

DATA = RAW

INFORMATION = PROCESSED + MEANINGFUL

10. In Computer Systems

Now connect this to the previous lectures.

We learned:

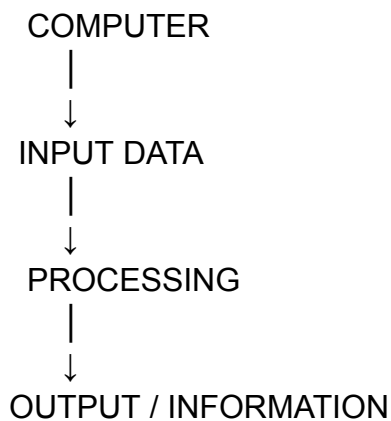
INPUT → PROCESSING → OUTPUT

And now:

DATA → PROCESSING → INFORMATION

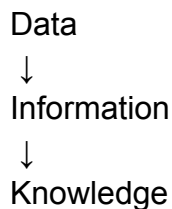
These ideas are closely connected.

For example:



11. Data → Information → Knowledge

You may encounter another concept in BSCS:



For example:

Data

80, 90, 70, 95, 85

Information

Average = 84

Highest = 95

Lowest = 70

Knowledge

From this information, a teacher might understand:

The student's overall performance is strong, but there is room for improvement in the subject where the 70 was obtained.

You don't need to memorize this third level yet. Just understand the progression.



The Core Concept

Imagine you work in a university.

The database contains:

Student ID

Name

Marks

Attendance

Department

These are **data**.

The university system processes them and produces:

Students eligible for scholarship

Students with low attendance

Students who passed

Department average

Class average

These are **information**.

The information helps people make decisions.
